

# INFINITY WAVE

## AI INTERNS LEARNING SERIES — WORKSHOP #02

# DATA PREPROCESSING

### Practical Intern Tasks

#### Workshop #02: Data Preprocessing — Preparing Data for Machine Learning

##### Objective:

Learn how to identify data-quality problems, make justified cleaning decisions, and transform categorical information into machine-readable representations.

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## TASK 1 — CLEAN THE DATA

### Scenario: E-Commerce Customer Dataset

You are working as a **Junior Data Engineer** at an e-commerce company.

The company wants to build a **Machine Learning model that predicts whether a customer is likely to make another purchase**.

Your team has received the customer dataset below.

Before the ML team can train a model, **your responsibility is to identify and handle data-quality problems**.

#### Your Mission

You have **7 minutes**.

For each problem you find:

1. Identify **what is wrong**.
2. Decide what action you would take:
3. **Remove**
4. **Correct**

5. **Impute / Fill**
6. **Investigate Further**
7. Give a **one-line reason** for your decision.

## Important

Do **not** assume that every unusual value is automatically wrong.

Think like a real data professional:

**Is it wrong? → Can we verify it? → What is the safest action?**

## TASK 1 DATA

| Customer ID | Name        | Age | City      | Purchase Amount | Email            |
|-------------|-------------|-----|-----------|-----------------|------------------|
| C001        | Ali Khan    | 24  | Lahore    | 4500            | ali@gmail.com    |
| C002        | Sara Ahmed  | 29  | Karachi   | 6200            | sara@gmail.com   |
| C003        | Hamza Malik | —   | Lahore    | 3800            | hamza@gmail.com  |
| C004        | Ayesha Noor | 26  | Islamabad | 5100            | ayesha@gmail.com |
| C002        | Sara Ahmed  | 29  | Karachi   | 6200            | sara@gmail.com   |
| C005        | Bilal Shah  | 245 | Lahore    | 4100            | bilal@gmail.com  |
| C006        | Hina Ali    | 31  | —         | 7200            | hina@gmail.com   |
| C007        | Usman Raza  | 27  | Karachi   | -500            | usman@gmail.com  |
| C008        | Ahmed Khan  | 22  | Lahore    | 3900            | —                |
| C009        | Zara Iqbal  | 28  | Lahore    | 4600            | zara@gmail.com   |
| C010        | Mariam Noor | 35  | Islamabad | 8100            | mariam@gmail.com |
| C011        | Danish Ali  | 19  | Karachi   | 2500            | danish@gmail.com |
| C012        | Sana Riaz   | —   | Lahore    | 5400            | sana@gmail.com   |
| C013        | Omer Farooq | 41  | Multan    | 9200            | omer@gmail.com   |
| C014        | Laiba Khan  | 23  | Lahore    | 4700            | laiba@gmail.com  |
| C015        | Fahad Ahmed | 33  | Karachi   | 6600            | fahad@gmail.com  |
| C016        | Noor Fatima | 27  | —         | 5300            | noor@gmail.com   |
| C017        | Hassan Raza | -4  | Lahore    | 4200            | hassan@gmail.com |
| C018        | Iqra Malik  | 30  | Islamabad | 6100            | iqra@gmail.com   |

| Customer ID | Name          | Age | City      | Purchase Amount | Email                 |
|-------------|---------------|-----|-----------|-----------------|-----------------------|
| C019        | Talha Shah    | 25  | Karachi   | 4800            | talha@gmail.com       |
| C020        | Eman Ali      | 21  | Lahore    | 3500            | eman@gmail.com        |
| C021        | Arham Khan    | 38  | Peshawar  | 7500            | arham@gmail.com       |
| C022        | Mahnoor Ali   | 26  | Karachi   | 5800            | mahnoor@gmail.com     |
| C023        | Saad Ahmed    | 32  | Lahore    | 6900            | saad@gmail.com        |
| C024        | Hoorain Raza  | 24  | Islamabad | 5100            | hoorain@gmail.com     |
| C025        | Adeel Malik   | 29  | Karachi   | 6200            | adeel@gmail.com       |
| C026        | Mehak Noor    | 22  | Lahore    | 4400            | mehak@gmail.com       |
| C027        | Hamza Malik   | 30  | Lahore    | 5700            | hamza2@gmail.com      |
| C028        | Usman Khan    | 45  | Quetta    | 9800            | usman.khan@gmail.com  |
| C029        | Areeba Shah   | 20  | Karachi   | 3200            | areeba@gmail.com      |
| C030        | Bilal Ahmed   | 34  | Lahore    | 7100            | bilal.ahmed@gmail.com |
| C031        | Sara Noor     | 27  | Islamabad | 6000            | sara.noor@gmail.com   |
| C032        | Waleed Ali    | 39  | Karachi   | 8300            | waleed@gmail.com      |
| C033        | Maha Khan     | 23  | Lahore    | 3900            | maha@gmail.com        |
| C034        | Rayan Ahmed   | 28  | Multan    | 5200            | rayan@gmail.com       |
| C035        | Anaya Raza    | 31  | Karachi   | 6800            | anaya@gmail.com       |
| C036        | Huzaifa Malik | 26  | Lahore    | 4600            | huzaifa@gmail.com     |
| C037        | Saba Ali      | 36  | Islamabad | 7900            | saba@gmail.com        |
| C038        | Zain Khan     | 24  | Karachi   | 4100            | zain@gmail.com        |
| C039        | Amna Farooq   | 29  | Lahore    | 5500            | amna@gmail.com        |
| C040        | Rafay Shah    | 33  | Peshawar  | 6400            | rafay@gmail.com       |

## TASK 1 — QUESTIONS

### Question 1

Find **all missing values** in the dataset.

For each one, decide whether you would:

- Remove the row
- Impute the value
- Leave it missing
- Investigate further

Explain why.

## Question 2

Find any **duplicate records**.

Should the duplicate be removed? Why?

## Question 3

Find any **invalid or impossible values**.

Explain why they are suspicious.

## Question 4

Find any **unusual values that should be investigated rather than immediately deleted**.

## Question 5

If this dataset were going into an ML model, what other data-quality checks would you perform?

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# TASK 1 — RESPONSE FORMAT

Use this structure:

| Row / Customer | Problem     | Action               | Reason                        |
|----------------|-------------|----------------------|-------------------------------|
| Example        | Missing Age | Impute / Investigate | Age is required for the model |
|                |             |                      |                               |
|                |             |                      |                               |
|                |             |                      |                               |
|                |             |                      |                               |
|                |             |                      |                               |

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# TASK 1 — INSTRUCTOR DISCUSSION

## Expected Problems

### C002 — Duplicate

Sara Ahmed appears twice with identical information.

**Suggested action:** Remove duplicate.

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### C003 — Missing Age

Age is missing.

**Possible action:** Investigate or impute using an appropriate strategy.

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### C005 — Age = 245

This is highly suspicious for a customer age.

**Possible action:** Investigate and correct if the source confirms an entry error.

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### C006 — Missing City

City is missing.

**Possible action:** Investigate or impute depending on how the feature will be used.

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### C007 — Purchase Amount = -500

This is unusual but should **not automatically be deleted**.

It could represent:

- A refund
- A reversed transaction
- A data-entry error

**Best first action:** Investigate.

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### **C008 — Missing Email**

Email is missing.

Whether this row should be removed depends on whether Email is actually required for the ML task.

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### **C012 — Missing Age**

Another missing value requiring an appropriate strategy.

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### **C016 — Missing City**

Another missing categorical value.

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### **C017 — Age = -4**

Impossible customer age.

**Suggested action:** Investigate and correct/remove depending on the source.

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## **KEY LESSON FROM TASK 1**

**Data cleaning is not simply deleting anything that looks strange.**

A good data professional asks:

**What is wrong?**

↓

**Why is it wrong?**

↓

**Can we verify it?**

↓

**What is the safest action?**

**Remember:**

**Garbage In → Garbage Out**

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## TASK 2 — TRANSFORM THE DATA

### Scenario: Cricket Performance Dataset

You are working on an **AI-powered cricket analytics system**.

The system will use player information to support player-performance analysis.

The ML team has received the dataset below.

The problem is that much of the information is **categorical and human-readable**.

A machine-learning algorithm may require these features to be represented in a suitable numerical form.

Your responsibility is to **transform the data into a representation that a machine-learning model can work with**.

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## YOUR MISSION

You have **10 minutes**.

### Step 1 — Classify the Data

Identify which columns are:

**Categorical**

and which are:

**Numerical / Identifier**

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### Step 2 — Identify Features That Need Transformation

Consider:

- Country
- Format

- Batting Style
  - Bowling Type
  - Role
- 

### Step 3 — Choose an Encoding Strategy

Decide how you would represent the categorical variables.

Consider:

- Label Encoding
  - One-Hot Encoding
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### Step 4 — Perform One-Hot Encoding

Create a one-hot representation for:

**Format**

and

**Role**

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### Step 5 — Explain Your Decision

Answer:

**Why shouldn't we blindly encode Pakistan = 1, India = 2, Australia = 3?**

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## TASK 2 DATA

| ID   | Player          | Country   | Format | Batting Style | Bowling Type | Role         |
|------|-----------------|-----------|--------|---------------|--------------|--------------|
| P001 | Babar Azam      | Pakistan  | ODI    | Right-hand    | None         | Batter       |
| P002 | Mohammad Rizwan | Pakistan  | T20    | Right-hand    | None         | Wicketkeeper |
| P003 | Steve Smith     | Australia | Test   | Right-hand    | None         | Batter       |
| P004 | Virat Kohli     | India     | ODI    | Right-hand    | None         | Batter       |

| ID   | Player                | Country      | Format | Batting Style | Bowling Type  | Role         |
|------|-----------------------|--------------|--------|---------------|---------------|--------------|
| P005 | Shaheen Afridi        | Pakistan     | ODI    | Left-hand     | Fast          | Bowler       |
| P006 | Rashid Khan           | Afghanistan  | T20    | Right-hand    | Leg Spin      | Bowler       |
| P007 | Joe Root              | England      | Test   | Right-hand    | None          | Batter       |
| P008 | Jasprit Bumrah        | India        | Test   | Right-hand    | Fast          | Bowler       |
| P009 | David Warner          | Australia    | T20    | Left-hand     | None          | Batter       |
| P010 | Kane Williamson       | New Zealand  | Test   | Right-hand    | None          | Batter       |
| P011 | Shadab Khan           | Pakistan     | T20    | Right-hand    | Leg Spin      | All-rounder  |
| P012 | Ben Stokes            | England      | Test   | Left-hand     | Medium        | All-rounder  |
| P013 | Trent Boult           | New Zealand  | ODI    | Left-hand     | Fast          | Bowler       |
| P014 | Quinton de Kock       | South Africa | T20    | Left-hand     | None          | Wicketkeeper |
| P015 | Rohit Sharma          | India        | ODI    | Right-hand    | None          | Batter       |
| P016 | Pat Cummins           | Australia    | Test   | Right-hand    | Fast          | Bowler       |
| P017 | Jos Buttler           | England      | T20    | Right-hand    | None          | Wicketkeeper |
| P018 | Mohammad Nabi         | Afghanistan  | T20    | Right-hand    | Off Spin      | All-rounder  |
| P019 | Babar Azam            | Pakistan     | T20    | Right-hand    | None          | Batter       |
| P020 | Rassie van der Dussen | South Africa | ODI    | Right-hand    | None          | Batter       |
| P021 | Marnus Labuschagne    | Australia    | Test   | Right-hand    | None          | Batter       |
| P022 | Ravindra Jadeja       | India        | Test   | Left-hand     | Left-arm Spin | All-rounder  |
| P023 | Haris Rauf            | Pakistan     | T20    | Right-hand    | Fast          | Bowler       |
| P024 | Shakib Al Hasan       | Bangladesh   | ODI    | Left-hand     | Left-arm Spin | All-rounder  |
| P025 | Kagiso Rabada         | South Africa | Test   | Right-hand    | Fast          | Bowler       |
| P026 | Tom Latham            | New Zealand  | ODI    | Left-hand     | None          | Wicketkeeper |
| P027 | Adil Rashid           | England      | ODI    | Right-hand    | Leg Spin      | Bowler       |

| ID   | Player            | Country      | Format | Batting Style | Bowling Type | Role         |
|------|-------------------|--------------|--------|---------------|--------------|--------------|
| P028 | Fakhar Zaman      | Pakistan     | ODI    | Left-hand     | None         | Batter       |
| P029 | Nicholas Pooran   | West Indies  | T20    | Left-hand     | None         | Wicketkeeper |
| P030 | Wanindu Hasaranga | Sri Lanka    | T20    | Right-hand    | Leg Spin     | All-rounder  |
| P031 | Litton Das        | Bangladesh   | ODI    | Right-hand    | None         | Wicketkeeper |
| P032 | Mustafizur Rahman | Bangladesh   | T20    | Left-hand     | Fast         | Bowler       |
| P033 | Glenn Maxwell     | Australia    | ODI    | Right-hand    | Off Spin     | All-rounder  |
| P034 | Suryakumar Yadav  | India        | T20    | Right-hand    | None         | Batter       |
| P035 | Imam-ul-Haq       | Pakistan     | ODI    | Left-hand     | None         | Batter       |
| P036 | Kusal Mendis      | Sri Lanka    | T20    | Right-hand    | None         | Wicketkeeper |
| P037 | Mark Wood         | England      | Test   | Right-hand    | Fast         | Bowler       |
| P038 | Aiden Markram     | South Africa | ODI    | Right-hand    | Medium       | All-rounder  |
| P039 | Ibrahim Zadran    | Afghanistan  | ODI    | Right-hand    | None         | Batter       |
| P040 | Shai Hope         | West Indies  | ODI    | Right-hand    | None         | Wicketkeeper |

## TASK 2 — PART A

### CLASSIFY THE COLUMNS

Complete the table.

| Column        | Categorical? | Numerical / Identifier? | Why? |
|---------------|--------------|-------------------------|------|
| Player        |              |                         |      |
| Country       |              |                         |      |
| Format        |              |                         |      |
| Batting Style |              |                         |      |
| Bowling Type  |              |                         |      |
| Role          |              |                         |      |

## TASK 2 — PART B

### CHOOSE THE TRANSFORMATION

For each categorical feature, decide whether you would use:

**Label Encoding**

or

**One-Hot Encoding**

| Feature       | Your Choice | Why? |
|---------------|-------------|------|
| Country       |             |      |
| Format        |             |      |
| Batting Style |             |      |
| Bowling Type  |             |      |
| Role          |             |      |

## TASK 2 — PART C

### ONE-HOT ENCODING

Use the following four players:

| Player  | Format | Role         |
|---------|--------|--------------|
| Babar   | ODI    | Batter       |
| Shaheen | ODI    | Bowler       |
| Rizwan  | T20    | Wicketkeeper |
| Rashid  | T20    | Bowler       |

Create the following columns:

**Format**

- ODI
- T20

- Test

## Role

- Batter
- Bowler
- Wicketkeeper
- All-rounder

Complete:

| Player  | ODI | T20 | Test | Batter | Bowler | Wicketkeeper | All-rounder |
|---------|-----|-----|------|--------|--------|--------------|-------------|
| Babar   |     |     |      |        |        |              |             |
| Shaheen |     |     |      |        |        |              |             |
| Rizwan  |     |     |      |        |        |              |             |
| Rashid  |     |     |      |        |        |              |             |

## TASK 2 — DISCUSSION CHALLENGE

Consider:

**Pakistan = 1**

**India = 2**

**Australia = 3**

**Question:**

**Would you use this representation directly for Country? Why or why not?**

Think carefully.

Does:

**India = 2**

mean that India is:

- twice as important as Pakistan?
- ranked above Pakistan?
- numerically greater than Pakistan?

No.

The numbers would be arbitrary category labels.

This is why encoding strategy matters.

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## TASK 2 — BONUS CHALLENGE

Choose **one additional categorical column** from the dataset.

For example:

### **Bowling Type**

Then decide:

1. What are its unique categories?
  2. How would you encode them?
  3. Why did you choose that method?
  4. Show the transformed representation for at least 4 players.
- 

## FINAL DISCUSSION

After completing both tasks, discuss:

### **Task 1 taught us:**

**Data must be cleaned before it can be trusted.**

### **Task 2 taught us:**

**Data must be represented in a form the model can use.**

Together:

**RAW DATA**

↓

**CLEAN**

↓

**HANDLE MISSING VALUES**



**TRANSFORM**



**PREPARE**



**TRAINING**

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## **WORKSHOP #02 KEY TAKEAWAY**

**"A powerful model cannot rescue bad data."**

**Remember:**

**Good AI starts with good data.**